

Human vs LLM Text Classification Using a Multi-Feature Approach

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SUMMARY OF THIS PAPER

A. Problem Setup

Advanced Large Language Models (LLMs) now produce text that is difficult to distinguish from human writing. This creates practical problems in areas requiring source verification, such as ensuring academic integrity and preventing the spread of AI-generated spam.

Our research develops a classification method utilizing various linguistic features such as lexical density, Personal Pronoun Proportion (PRP), and thematic features. We expect that a more accurate distinction is possible by combining multiple types of linguistic features simultaneously. Instead of relying on one signal, we use:

1. **Stylistic and Readability Features:** Metrics that describe *how* the text is written (e.g., sentence complexity, use of personal pronouns).
2. **Thematic Features:** Topic distributions derived from Latent Dirichlet Allocation (LDA) that describe *what* the text is about.

This leads to our primary research questions:

- Q1:** To what extent can the fundamental differences between human and LLM texts be quantified through a measurable set of linguistic features?
- Q2:** Can the proposed multi-feature representation effectively distinguish one LLM from others?

B. Novelty

The novelty of this work has two key contributions that, when combined, create our analytical framework:

- **A Deliberate Combination of Heterogeneous Features:** Our first contribution is the deliberate **combination** of two fundamentally different feature categories. We constructed a hybrid vector that integrates (1) a variety of stylistic metrics calculated using libraries like textstat and custom scripts for metrics such as the Personal Pronoun (PRP), with (2) thematic distributions derived from a Hierarchical Latent Dirichlet Allocation (LDA) model. This approach of fusing stylistic and thematic data into a single vector creates a richer feature set designed to capture a more holistic view of a text’s origin.
- **An Analytical Framework for Result Interpretation:** Our second contribution is the analytical framework itself. This approach moves beyond simply reporting accuracy scores and mandates a deeper analysis to explain why the features have potential to discriminate between LLMs and human. By this, we can identify the key linguistic fingerprints of each source. For instance, these analyses reveal that human sources consistently utilize a higher proportion of personal pronouns (PRP), reflecting a distinct narrative style.

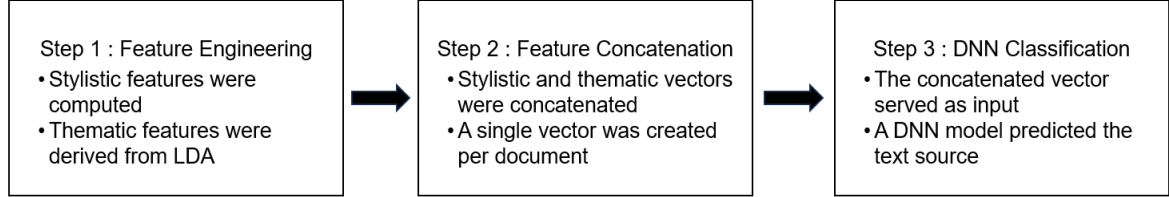
C. Algorithm

Data Collection and Preprocessing: Our study began with the careful construction of a corpus. The dataset comprises a total of 14,832 documents, sourced from 5 distinct sources: four LLMs and human sources. For text from human sources, we sourced data from Quora, ensuring all the answers were posted at least five years prior to mitigate potential LLM contamination. We identified approximately 100 unique questions and sampled up to 30 answers per question. To generate the LLM-generated text, we used the same set of questions as prompts for four LLM sources (Claude, DeepSeek, Gemini, GPT). The entire corpus then underwent a standardized preprocessing pipeline, including lowercasing, tokenization, and the removal of stopwords and special characters.

Our methodology is implemented as a systematic, three-step classification pipeline.

Model Architecture: For model selection, we tried three architectures, including tree-based ensembles (LightGBM) and TabNet, an attention-based architecture. A Multi-Layer Perceptron (MLP) was ultimately selected for its highest performance on our dataset. The chosen network consists of four sequential hidden layers (256, 128, 64, 32), with Batch Normalization and Dropout applied after each layer.

The preprocessed data was passed through the following three-step pipeline:



D. Experiments

To validate the performance of our algorithm, we conducted a classification experiment with a hold-out validation. The dataset of 14,832 documents was randomly partitioned into training (80%) and test (20%) sets. The DNN model, trained on the comprehensive hybrid features, was rigorously evaluated on the test set and achieved an overall 82.2% accuracy. The detailed results are presented in the figures below. As shown in the confusion matrix (Figure 1), the model demonstrated high performance, notably classifying human-sourced text with the highest accuracy (F1-Score: 0.902) and differentiating among LLMs with high accuracy. The boxplot of the PRP ratio (Figure 2) reveals human texts exhibit a consistently higher PRP ratio which serves as a useful feature for classification. Our analysis further shows that features contribute in distinct ways: the PRP ratio is particularly effective for discriminating between human and LLM texts, while lexical density shows its strength in classifying among the different LLM sources. And LDA is helpful at all sources. Furthermore, an ablation study was conducted. The model trained only on thematic (LDA) features achieved a notable 63.2% accuracy. The combination of thematic features and PRP achieves 78.9% accuracy. Finally, the use of thematic features, PRP, and lexical density achieves 81.2% accuracy.

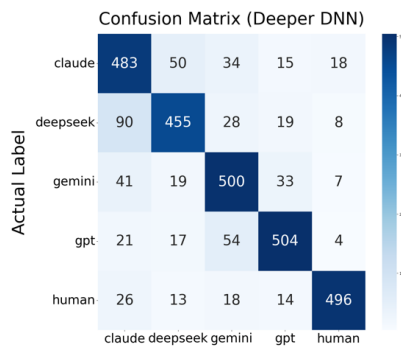


Figure 1: Confusion Matrix

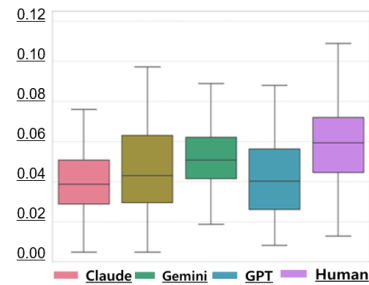


Figure 2: Personal Pronoun Proportion

Our study successfully demonstrates a framework for classifying texts generated purely by either humans or LLMs. However, more features should be explored for the classifying LLM and human. Real world scenarios increasingly involve mixed-authorship texts (mixtexts), where human writing is revised by AI or vice versa. As highlighted by recent research (e.g., Zhang et al., 2024), detecting blended texts poses a significant challenge.